

IBM – Agentic AI

Assignment – 2

AI-Powered Customer Support Automation System

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Output Screenshots

Entry Page:

```
python3
> python3 customer_support_system.py --fresh
[Setup] Initializing ChatOllama LLM (qwen2.5:3b)...
[Setup] --fresh flag: memory.db cleared. Starting with clean memory.
[RAG] Initializing RAG pipeline (ChromaDB persistent store)...
[RAG] Loading existing ChromaDB index from './chroma_db/' (KB unchanged).
[RAG] Loaded 21 chunks from persistent vector store. Ready.

[Setup] Created new SQLite memory database: 'memory.db'
[Setup] Building LangGraph workflow...
[Setup] Initializing ChatOllama LLM (qwen2.5:3b)...
[Setup] LangGraph workflow compiled successfully.
[Setup] Workflow diagram saved as 'workflow_diagram.png'

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ABC Technologies Pvt. Ltd.
AI-Powered Customer Support Automation System
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LLM      : Ollama qwen2.5:3b
Embeddings : nomic-embed-text (ChromaDB)
Memory   : SQLite (memory.db) [persistent across sessions]
Knowledge : company_policy | pricing_guide | technical_manual | faq

=====
Tip: Use --fresh flag to wipe memory and start a clean session.

Enter Customer ID (press Enter for 'cust_001'): cust1
Enter your name (press Enter to skip): Madhan

New session started for Customer ID: 'cust1'.
Type your query below. Type 'exit' or 'quit' to end.
```

Query 1:

What are the pricing plans available for your software?

```
python3

Tip: Use --fresh flag to wipe memory and start a clean session.

Enter Customer ID (press Enter for 'cust_001'): cust1
Enter your name (press Enter to skip): Madhan

New session started for Customer ID: 'cust1'.
Type your query below. Type 'exit' or 'quit' to end.

Customer > What are the pricing plans available for your software?

Customer [cust1] > 'What are the pricing plans available for your software?'

[Memory] New customer 'cust1' - starting fresh session.
[Graph] Running...

[Node: classify_intent] Classifying: 'What are the pricing plans available for your software?'
[Node: classify_intent] Category: Sales | Reason: The query is about the pricing plans of the software.
[Node: retrieve_context] Searching knowledge base for: 'What are the pricing plans available for your software?'
[Node: retrieve_context] Retrieved 3 relevant chunks.
[Node: sales_agent] Running Sales Support agent...
[Node: sales_agent] Draft response: 'We offer three subscription plans tailored to different business sizes and needs. Here's a quick ove...'
[Node: check_risk] Low-risk request. No human approval needed.
[Node: supervisor_validation] Polishing the drafted response...
[Node: supervisor_validation] Final response polished and ready.

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FINAL RESPONSE TO CUSTOMER
=====

We offer three subscription plans tailored to different business sizes and needs. Here's a quick overview:

- For small businesses, we have the **Starter Plan** at ₹2,499 per month (or ₹24,990 annually), with up to 5 users and basic features like task boards and secure cloud storage on AWS Mumbai region.
- If you're looking for more advanced tools, our **Growth Plan** is available at ₹6,999 per month (or ₹69,990 annually) for up to 25 users. It includes everything from the Starter plan plus additional features such as custom reports and workflow automations.
- Our **Free Trial & Sandbox** offers a 14-day free trial on both Starter and Growth plans with priority email and live chat support.

Would you like more details on any of these plans?

Customer > █
```

Query 2:

I forgot my account password.

```
python3
[Node: supervisor_validation] Final response polished and ready.

FINAL RESPONSE TO CUSTOMER

We offer three subscription plans tailored to different business sizes and needs. Here's a quick overview:

- For small businesses, we have the **Starter Plan** at ₹2,499 per month (or ₹24,990 annually), with up to 5 users and basic features like task boards and secure cloud storage on AWS Mumbai region.
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- Our **Free Trial & Sandbox** offers a 14-day free trial on both Starter and Growth plans with priority email and live chat support.

Would you like more details on any of these plans?

Customer > I forgot my account password.

Customer [cust1] > 'I forgot my account password.'

[Memory] Returning customer 'cust1' - 2 prior message(s) loaded from SQLite.
[Graph] Running...

[Node: classify_intent] Classifying: 'I forgot my account password.'
[Node: classify_intent] Category: Account | Reason: Customer requested password reset for their account.
[Node: retrieve_context] Searching knowledge base for: 'I forgot my account password.'
[Node: retrieve_context] Retrieved 3 relevant chunks.
[Node: account_agent] Running Account Support agent...
[Node: account_agent] Draft response: 'Hello Madhan, I understand you need to reset your password. To proceed, please click the "Forgot Password?" link on the login page. Your new password must be at least 12 characters long and include uppercase letters (A-Z), lowercase letters (a-z), digits (0-9), and special characters such as !, @, #, $, %, etc.'
[Node: check_risk] Low-risk request. No human approval needed.
[Node: supervisor_validation] Polishing the drafted response...
[Node: supervisor_validation] Final response polished and ready.

FINAL RESPONSE TO CUSTOMER

Hello Madhan, I understand you need to reset your password. To proceed, please click the "Forgot Password?" link on the login page. Your new password must be at least 12 characters long and include uppercase letters (A-Z), lowercase letters (a-z), digits (0-9), and special characters such as !, @, #, $, %, etc.

If you need further assistance with this process or have any other account-related queries, feel free to ask.

Customer > |
```

Query 3:

My application crashes whenever I upload a file.

```
python3
[Memory] Returning customer 'cust1' - 4 prior message(s) loaded from SQLite.
[Graph] Running...

[Node: classify_intent] Classifying: 'My application crashes whenever I upload a file.'
[Node: classify_intent] Category: Technical Support | Reason: Application crashes whenever I upload a file.
[Node: retrieve_context] Searching knowledge base for: 'My application crashes whenever I upload a file.'
[Node: retrieve_context] Retrieved 3 relevant chunks.
[Node: technical_agent] Running Technical Support agent...
[Node: technical_agent] Draft response: 'Hello Madhan,

I understand that your application is crashing when you try to upload files. Let's go...'
[Node: check_risk] Low-risk request. No human approval needed.
[Node: supervisor_validation] Polishing the drafted response...
[Node: supervisor_validation] Final response polished and ready.

FINAL RESPONSE TO CUSTOMER

Hello Madhan,

I understand that your application is crashing when you try to upload files. Let's go through some troubleshooting steps together to resolve the issue:

1. **Check File Size**: Ensure that the file you are trying to upload does not exceed the size limit for your plan. For the Starter Plan, the maximum file size is 50 MB. Please confirm if this matches what you have set up.
2. **Verify File Format**: Confirm that the file format you are uploading is one of the supported formats: PDF, DOCX, XLSX, CSV, PNG, or JPG. If it's not in a supported format, please convert the file to one of these types before attempting to upload again.
3. **Clear Browser Cache and Cookies**:
   - Press 'Ctrl + Shift + Delete' on your keyboard.
   - Select "Cookies and other site data" and "Cached images and files".
   - Click "Delete" to clear them.
   - Try opening the application in an incognito window or private browsing mode.
4. **Switch to a Stable Internet Connection**:
   - If you are using a mobile hotspot, try switching to a stable broadband connection such as BSNL, Jio Fiber, ACT, or Airtel Xstream for better performance.
5. **Check Console Logs (if the issue persists)**:
   - Open your browser's developer tools by pressing 'F12'.
   - Navigate to the "Console" tab.
   - If there are any error messages displayed, please share them with our support team so we can further diagnose and resolve the issue.

If you have tried these steps and the application is still crashing when uploading files, it would be helpful if you could provide more details about your current setup or any additional error messages that might appear in the console. We will work together to find a solution.

Thank you for your patience while we troubleshoot this issue.

Customer > |
```

Query 4:

I need a refund for my annual subscription.

```
python3
Customer > I need a refund for my annual subscription.

Customer [cust1] > 'I need a refund for my annual subscription.'
[Memory] Returning customer 'cust1' - 6 prior message(s) loaded from SQLite.
[Graph] Running...

[Node: classify_intent] Classifying: 'I need a refund for my annual subscription.'
[Node: classify_intent] Category: Billing | Reason: Refund request related to annual subscription
[Node: retrieve_context] Searching knowledge base for: 'I need a refund for my annual subscription.'
[Node: retrieve_context] Retrieved 3 relevant chunks.
[Node: billing_agent] Running Billing Support agent...
[Node: billing_agent] Draft response: 'Hello Madhan,

Thank you for reaching out regarding your refund request. Under our policy, you are e...'
[Node: check_risk] High-risk detected! Reason: Query contains high-risk keyword: 'refund'

=====
HUMAN SUPERVISOR APPROVAL REQUIRED
=====
Customer Name : Madhan
Category      : Billing
Query         : I need a refund for my annual subscription.

Draft Response:
Hello Madhan,

Thank you for reaching out regarding your refund request. Under our policy, you are eligible for a full refund within 30 days of the initial purchase date if you are unsatisfied with the product. Your refund will be processed in accordance with our refund policy.

Here's what to expect:

- **Refund Policy**: We offer a full refund if requested within 30 days of your initial purchase, provided that all terms and conditions have been met.
- **Review Process**: All refund requests are reviewed by a human supervisor before processing. This ensures the integrity and compliance with RBI guidelines.

To proceed with your request, please submit it through our support portal or contact Billing Support directly. Once submitted, we will review your request within 5-7 business days (excluding Indian bank holidays and public holidays).

Please provide us with your original purchase date so that we can accurately process the refund for you.

If you have any further questions or need additional assistance, feel free to ask. We are here to help.

Thank you for choosing ABC Technologies.
```

```
python3
Here's what to expect:

- **Refund Policy**: We offer a full refund if requested within 30 days of your initial purchase, provided that all terms and conditions have been met.
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Please provide us with your original purchase date so that we can accurately process the refund for you.

If you have any further questions or need additional assistance, feel free to ask. We are here to help.

Thank you for choosing ABC Technologies.

=====
Approve this request? (y/n): y
[Supervisor] Request APPROVED.

[Graph] Resuming after supervisor decision...
[Node: supervisor_validation] Polishing the drafted response...
[Node: supervisor_validation] Final response polished and ready.

=====
FINAL RESPONSE TO CUSTOMER
=====
Hello Madhan,

Thank you for reaching out regarding your refund request. Under our policy, you are eligible for a full refund within 30 days of the initial purchase date if you are unsatisfied with the product. Your refund will be processed in accordance with our refund policy.

Here's what to expect:

- **Refund Policy**: We offer a full refund if requested within 30 days of your initial purchase, provided that all terms and conditions have been met.
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If you have any further questions or need additional assistance, feel free to ask. We are here to help.

Thank you for choosing ABC Technologies.

Customer > 
```

Query 5: (Works cross sessions too – matches with custId)

I need a refund for my annual subscription.

```
python3
LLM      : Ollama qwen2.5:3b
Embeddings : nomic-embed-text (ChromaDB)
Memory    : SQLite (memory.db) [persistent across sessions]
Knowledge  : company_policy | pricing_guide | technical_manual | faq

Tip: Use --fresh flag to wipe memory and start a clean session.

Enter Customer ID (press Enter for 'cust_001'): cust1
Enter your name (press Enter to skip): Madhan

Welcome back! Found 13 prior message(s) in memory for 'cust1'.
You can ask 'What was my previous support issue?' to recall history.
Type your query below. Type 'exit' or 'quit' to end.

Customer > What was my previous support issue?

Customer [cust1] > 'What was my previous support issue?'

[Memory] Returning customer 'cust1' – 13 prior message(s) loaded from SQLite.
[Graph] Running...

[Node: classify_intent] Classifying: 'What was my previous support issue?'
[Node: classify_intent] Category: Account | Reason: Customer asked about their previous support issues and account-related information.
[Node: memory_recall] Customer is asking about previous interactions.
[Node: memory_recall] Recall response generated from 7 prior customer queries.

FINAL RESPONSE TO CUSTOMER

Hello Madhan,

Here are your previous support topics:
1. Pricing plans available for our software.
2. Forgot account password.
3. Application crashes when uploading files.
4. Requested a refund for annual subscription.

Let me know if there's anything else I can assist with.

Customer > █
```